Dataset Details

Supply Chain Disruption and Recovery Dataset

1. Overview

Item	Details
Dataset name	Supply Chain Disruption and Recovery Dataset
Dataset source	Kaggle (public dataset, as specified in the project PRD)
Dataset link	https://www.kaggle.com/datasets/likithagedipudi/supply-chain-disruption-and-recovery-dataset
Data format	CSV (comma-separated, header row), UTF-8
File	data/supply_chain_disruption_recovery.csv (sample: data/sample_data/sample_disruptions.csv)
Size	100,000 rows x 15 columns, 17.7 MB in memory
Missing values	0 | duplicate IDs: 0

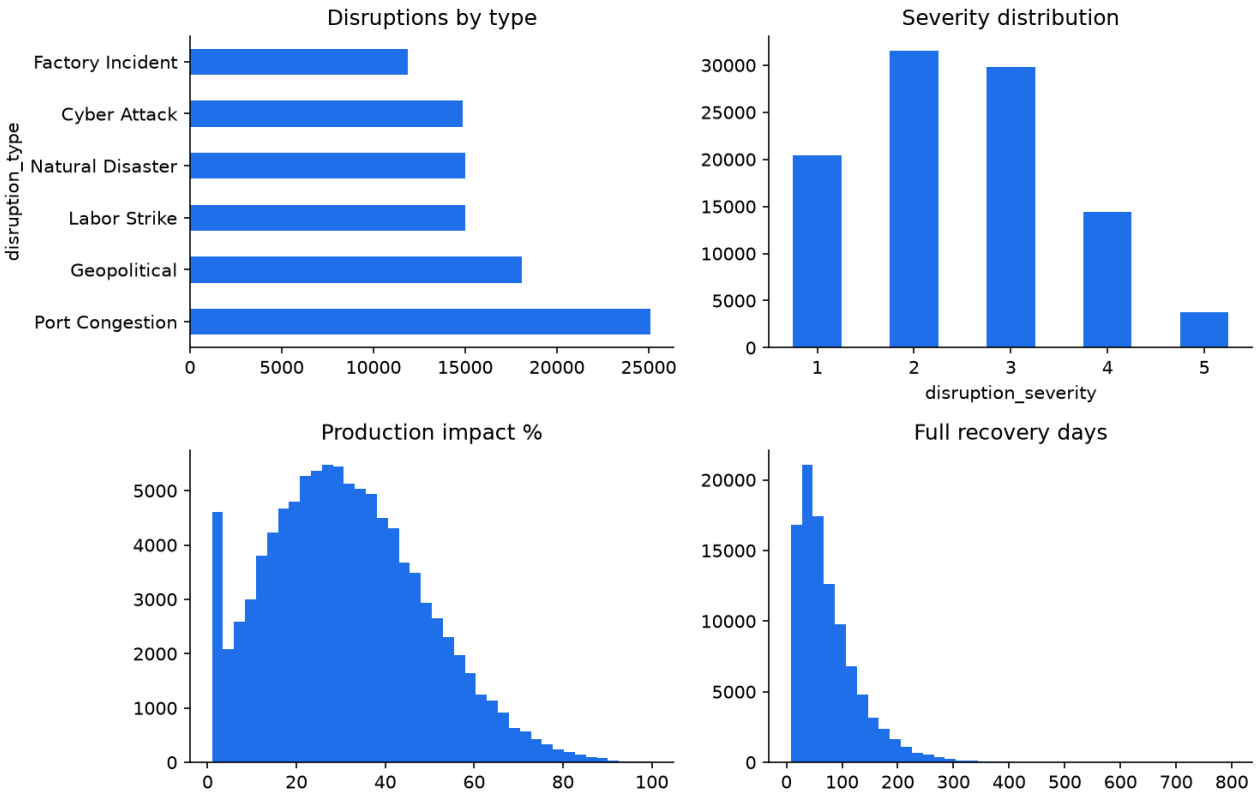
2. Dataset Purpose

Each row is a historical supply chain disruption event with the affected supplier's characteristics, the disruption's severity and impact, the response strategy taken and how long recovery took. In this project the dataset is the knowledge base for retrieval (finding similar past disruptions), the training data for impact, loss, recovery-time and risk prediction models, and the evidence base for response recommendations, prioritization calibration, explainability and evaluation ground truth.

The dataset is fully structured: every field is a category, a number or a yes/no flag, with no free-text descriptions. The narratives that make retrieval possible are generated from the fields (section 4).

3. Key Fields

Field	Type	Description	Values / range
disruption_id	str	Unique event identifier (DIS-######)	100,000 unique, e.g. DIS-000001
disruption_type	str	Cause category of the disruption	Port Congestion (25,118), Geopolitical (18,084), Labor Strike (15,017), Natural Disaster (15,016), Cyber Attack (14,887), Factory Incident (11,878)
industry	str	Industry of the affected supply chain	Electronics (29,844), Automotive (24,934), Consumer Goods (20,158), Pharmaceuticals (15,015), Aerospace (10,049)
supplier_tier	int64	Supplier tier (1 = direct supplier, 4 = deep sub-tier)	1 (34,938), 2 (30,023), 3 (25,019), 4 (10,020)
supplier_region	str	Geographic region of the supplier	Asia-Pacific (40,262), Europe (21,890), North America (19,939), South America (9,896), Africa/Middle East (8,013)
supplier_size	str	Supplier company size	Medium (45,161), Small (29,872), Large (24,967)
has_backup_supplier	bool	Whether a backup supplier existed	True (55,574), False (44,426)
disruption_severity	int64	Severity on a 1 (minor) - 5 (critical) scale	1 (20,390), 2 (31,553), 3 (29,804), 4 (14,462), 5 (3,791)
production_impact_pct	float64	Production capacity lost (%)	min 1.0 · median 30.1 · mean 31.4 · max 100.0
revenue_loss_usd	float64	Financial loss (USD)	min 10,000.0 · median 875,466.2 · mean 2,522,811.9 · max 373,130,977.5
response_type	str	Response strategy applied	Alternative Supplier (44,362), Production Reroute (18,858), Inventory Buffer (12,687), Customer Delay (12,429), Combined Strategy (11,664)
response_time_days	int64	Days until the response started	min 1.0 · median 2.0 · mean 4.1 · max 64.0
partial_recovery_days	int64	Days until partial recovery	min 3.0 · median 20.0 · mean 27.5 · max 337.0
full_recovery_days	int64	Days until full recovery	min 8.0 · median 61.0 · mean 75.8 · max 798.0
permanent_supplier_change	bool	Whether the supplier was permanently replaced	False (76,080), True (23,920)



4. Data Preprocessing Performed

- Deduplication on disruption_id and removal of rows missing an ID or disruption type (0 removed: the data is clean).
- Type casting: TRUE/FALSE strings to booleans; days, tier and severity to integers; impact and loss to floats; median imputation for any missing numerics.
- Range validation: production impact clipped to 0-100%, revenue loss >= 0, severity clipped to 1-5; category labels stripped of whitespace.
- Derived features: recovery_gap_days (full - partial), loss_bucket (low <P25, medium <P50, high <P90, critical), impact_score (mean of normalised impact, log-loss and recovery time), high_impact flag (top 25% impact_score, target of the risk model), response_effectiveness (0-100: recovery speed relative to cases of the same type & severity).
- Chunking: every record is converted to one natural-language narrative (one chunk per record) enriched with type synonyms, e.g. 'natural disaster such as earthquake, flood, typhoon...', so free-text queries match.
- Indexing: narratives embedded with all-MiniLM-L6-v2 into ChromaDB with all categorical fields as metadata (the vectors are also cached as a NumPy matrix for exact filtered search), and indexed with BM25 for keyword search.
- Model inputs: categorical fields as LightGBM categoricals; revenue loss modelled on log scale; 80/20 train/test split.
- Processed data cached as artifacts/processed.parquet; a 1,000-row stratified sample is saved to data/sample_data/.

Example chunk (record DIS-000001)

```
Moderate severity 2 Factory Incident (factory incident such as plant fire, explosion, equipment breakdown or quality c
    ontamination) affecting a Tier-1 (direct) large-sized Pharmaceuticals supplier in Asia-Pacific with no backup supp
    lier. Production dropped 54.3% with revenue loss of $11.38M (critical loss). Response: Customer Delay initiated af
    ter 14 days. Partial recovery in 3 days, full recovery in 59 days; the original supplier was retained.
```

Type descriptors used in chunks

Disruption type	Descriptor
Natural Disaster	natural disaster such as earthquake, flood, typhoon, hurricane, storm or wildfire
Cyber Attack	cyber attack such as ransomware, malware, hacking or IT systems outage
Labor Strike	labor strike, union walkout or workforce industrial action
Port Congestion	port congestion, shipping backlog, container and vessel delays, logistics bottleneck
Geopolitical	geopolitical event such as sanctions, tariffs, export bans, trade restrictions or conflict
Factory Incident	factory incident such as plant fire, explosion, equipment breakdown or quality contamination